Atlas 210X — VFO Nano add-on wiring

YT3CT (ex YU5KBM) · Nano ATmega328P (Old Bootloader) · sketch atlas_vfo_fixed.ino · 2026-10 · button / relay part = proposal

1. LCD RG1602A (16×2) — twisted wiring, 12 wires

Twisted group	Nano board pad	LCD pin
orange / black	orange: D8 → 470 Ω	6 E
	black: GND	1
blue / black	blue: D7 → 470 Ω	4 RS
	black: GND	1
red / black / yellow / grey	red: +5 V	2 VDD
	black: GND	1
	yellow: contrast pot wiper	3 V0
	grey: 10 Ω pad (backlight)	15 A
brown / white	brown: D9 → 470 Ω	11 D4
	white: D10 → 470 Ω	12 D5
purple / green	green: D11 → 470 Ω	13 D6
	purple: D12 → 470 Ω	14 D7

- **Nano board:** pads unchanged; all 3 black wires on the one GND pad. 470 Ω in line within ~2 cm of the pads (6 signal wires only).
- **At the LCD:** existing jumper 16–5–1 stays; 100 nF + 10 µF across pins 2–1. Pins 7–10 open.
- Different twist per group, < 1 cm untwisted at the ends. Groups side by side along the chassis.

2. TTL programming leads (no USB needed)

USB-TTL cable (Prolific)	Nano
Green (TX)	D0 / RX
White (RX)	D1 / TX
Black (GND)	GND
Red (+5 V)	**NOT connected**
—	RST lead (+ optional push button to GND)

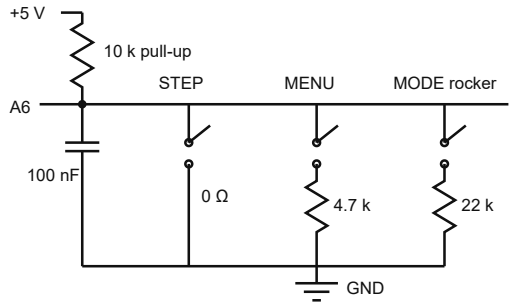
- No sync: swap green / white. **Upload:** start it, then RST to GND for ½ s (VariBeam Nano the same).
- Power cycle only with the POW LED fully out (TTL back-feeds the Nano). Never USB and TTL at the same time.

Free Nano pins & plan proposal

Pin	Now	Plan
D0 / D1	Serial	TTL / J4 serial
D4	free	AGC fast switch sense
D5	free	ATT relay sense
D6	STEP button	PRE relay sense (STEP → A6)
D13	free	CAT PTT → NPN (wire run)
A6	free (analog)	STEP, MENU + MODE rocker
A7	free (analog)	S3 position: CW / CAL sense

Used: D2/D3 encoder, D7–D12 LCD, A0/A1 S-meter, A3 band, A4/A5 I²C Si5351. **A2 TX** (LOW = TX): KEY line → 1N4148 (band to key line) → A2.
Sense inputs: GND-side switch + 1N4148, or 22 k / 10 k from +12 V. Parts at the **source end** (wires to the Nano carry ≤ 5 V). Never raw +13 V on a pin.

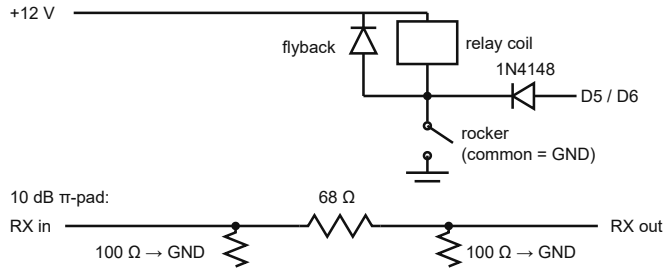
3. STEP / MENU buttons + MODE rocker on A6 proposal



Control	R to GND	ADC	Accept if
STEP button	0 Ω (direct)	0	< 120
MENU button	4.7 k	327 (285 with MODE)	120–480
MODE rocker (OPP)	22 k	703	480–860
nothing	—	1023	> 860

A6/A7 have **no internal pull-up** → the external 10 k is required. MODE is latching (NORM / OPP like the original S1): its state is taken while no button is pressed. 5 % resistors fine.

4. ATT / NORM / PRE — ON-OFF-ON rocker, DPDT relays, RX only proposal

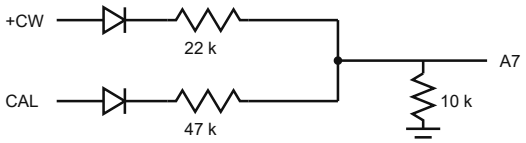


- Coil high end on +12 V, the rocker switches the coil low end to GND (ATT or PRE, centre = NORM). The 1N4148 (band to the coil) passes the LOW to the Nano pin and blocks the +12 V when off.
- DPDT signal relays: both poles switch the RF (in and out). π-pad 100 / 68 / 100 Ω ≈ 10 dB (exact: 96 / 71 / 96 Ω).
- **Only in the receive line after the antenna relay — never in the TX path.** Short leads / thin coax.

5. AGC fast/slow — rocker switch, sense D4 proposal

- 2N7000 at PC-300D: gate via 100 Ω from the rocker (+12 V), 100 k gate–GND, source GND, drain via R_{fast} (~150–330 k, to try) to the AGC hold node on PC-300D.
- Parts on PC-300D to be identified by value first (hold cap / 1 M discharge R).
- Sense: rocker on the GND side + 1N4148 to D4, or 22 k / 10 k divider from the switched +12 V.

6. S3 CW / CAL sense on A7 proposal



S3	A7 (13 V)	Nano
CW	≈ 3.8 V	BFO ~800 Hz into the passband, LO follows
CAL	≈ 2.1 V	CLK2 calibrator on
REC / TRANS	0 V	normal

MODE rocker NORM / OPP = sideband as the Atlas S1: software only (BFO to the other side of the crystal filter, 5.520.0 ↔ ~5.523.3 MHz, LO follows).

7. Front panel gain controls — 2 concentric dual pots, fixes fitted 2026-10

Pot	Function	Orig.	Fitted	Fix
1	AF gain (R4)	10 k	10 k	none
	RF gain (R6)	10 k	5 k	R5 4.7 k → 2.2 k, R7 470 → 220 Ω
2	MIC gain (R12)	10 k	20 k	20 k across outer lugs
	ALC (R11)	10 k	100 k	11 k across outer lugs

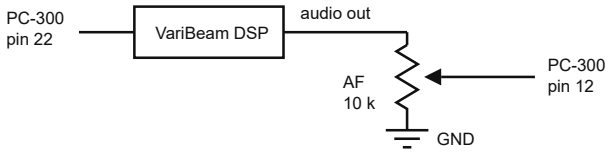
- RF gain: GND–R5–pot–R7–+8.5 V, wiper to PC-100 pin 14; wiper range ~2.6–8.2 V again. Measure R5 first (if already ~2.4 k, only R7).
- Resistor across the outer lugs = 10 k load as original, wiper ratio unchanged. 100 k ALC = harder, slower ALC → first suspect for low modulation.

8. J4 rear socket (old EXT OSC) — CAT / PTT / audio

proposal

Pin	Use
1	RST (grey) — **10 nF (103) in series** between pin 1 and Nano RST, at the Nano end (not to GND). From adapter DTR (programming from the rear)
2	Nano TX D1 (green) → adapter RX
3	Nano RX D0 (white) ← adapter TX
4	GND (black, chassis / Nano GND)
5	Audio in (PC → TX, via transformer, attenuated)
6	Audio GND, PC side — **floating**, open to chassis
7	Audio out (RX → PC, via transformer)
8	PTT (white) to the **KEY line** (mic PTT / PC-100 pin 20), grounded from the PC side. **Not the T line.**
9	spare

- Old wires (2–3 jumper, 5 +8.5 V, 6 stub, 7 T line, 8 +13 V fused) removed, GND on 4 kept. Check REC / SSB TX / CW after.
- USB-TTL adapter outside. Audio via 600:600 transformers, shielded, away from the serial pair. RX audio from PC-300 pin 22 (before the VariBeam).
- PTT now: PC grounds pin 8 (adapter RTS via NPN/opto). Later CAT PTT: D13 → 4.7 k → NPN base (BC547, 47 k B–E), collector to KEY line, emitter GND (KEY line draws 2.1 mA). PTT on RTS, never DTR (resets the Nano). Debug serial off with CAT.



- AGC / S-meter pickoff is before pin 22. Shielded audio leads, shield grounded at one end.
- R13 10 k (mic gain top) goes to the WHT/BLK **+CW** line (S3 CW / PC-100 pin 18), 0 V in SSB. Not PC-100 pin 21 (+13 V on every TX).

9. Calibrator — Si5351 CLK2 **proposal**

- Thin coax from CLK2 to where the PC-820 100 kHz cal board fed the receiver input — **run it now while the VFO box is open** (injection point to be located), receiver end left open for now.
- Inject weakly: 1–2 pF or 47 k in series at the injection point (marker about S9 or less).
- CLK2 on only in S3 CAL (A7 sense) or from the menu — off otherwise (a running CLK2 can add spurs).
- 100 kHz-style markers or an exact carrier (e.g. 14.100.000). With a GPSDO 1PPS the Nano can calibrate CORR automatically.

10. Next VFO Nano upload

- Si5351 spur fix: BFO on PLLB (integer), LO with an even integer output divider.
- CW: A7 sense, BFO shift ~800 Hz in CW, LO follows.
- Then re-check the 20 m birdie list (status notes).